

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S. No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	HDI Score Prediction	Takes 4 country indicators as input and predicts the HDI score using trained Linear Regression model	Flask backend — app.py / predict() route	Done	High
2	HDI Category Classification	Classifies predicted score into Low (0.30–0.40), Medium (0.40–0.70), High (0.70–0.80), Very High (0.80–0.94)	Flask backend — app.py if-elif logic	Done	High
3	Live HDI Score Estimator	Updates estimated HDI score in real time as user types values — no server call needed	indexnew.html — JavaScript estimateHDI() function	Done	High
4	Interactive Prediction Form	Dark themed web form with 4 input fields, range hints, progress bars under each field, and step tracker	indexnew.html — HTML/CSS/JavaScript	Done	High
5	Home Page	Introduction page explaining HDI concept, 4 category cards, features section and Predict Now button	home.html — HTML/CSS	Done	High
6	Result Page	Displays predicted HDI score, category, gradient scale bar, category description and action buttons	resultnew.html — HTML/CSS/Jinja2	Done	High

7	Data Visualization	Scatter plots for Schooling vs HDI and Life Expectancy vs HDI, correlation heatmap generated in notebook	HumDevIndex.ipynb — Matplotlib/Seaborn	Done	High
8	Model Persistence	Saves trained Linear Regression model as HDI.pkl using Pickle for reuse without retraining	HumDevIndex.ipynb + Flask/HDI.pkl	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home page, Interactive prediction form with live estimator, Result page with gradient scale
2	Backend / Logic	3	Flask routing, HDI score prediction, HDI category classification using if-elif
3	Database / Storage	1	HDI.csv dataset storage, HDI.pkl trained model saved using Pickle
4	API / Integration	1	Flask serves prediction results via /predict POST route to frontend HTML templates
5	Security / Authentication	0	No authentication required — open access web app for educational use

